

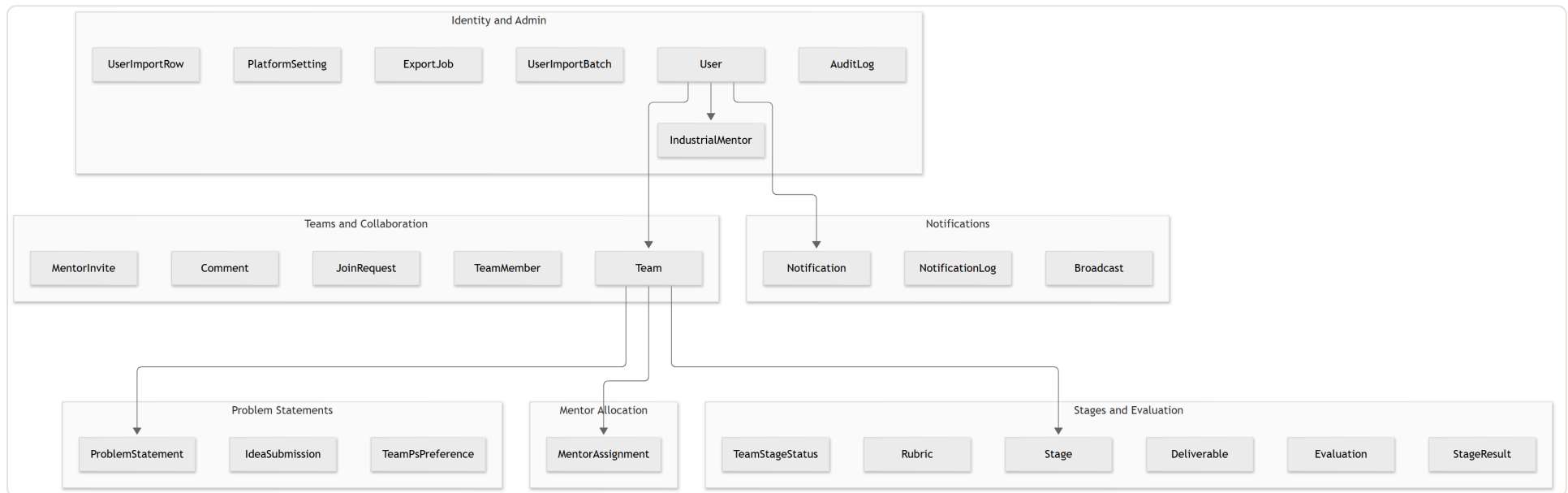
Prabodh — Entity Relationship Diagrams

Source: backend/prisma/schema.prisma · 25 PostgreSQL tables · JWT auth (no Clerk) · 2026-09-29

Schema note: Removed clerk_user_id, clerk_org_id, and clerk_invitation_id. Teams use team_code; users authenticate with email + password/OTP and app-issued JWT.

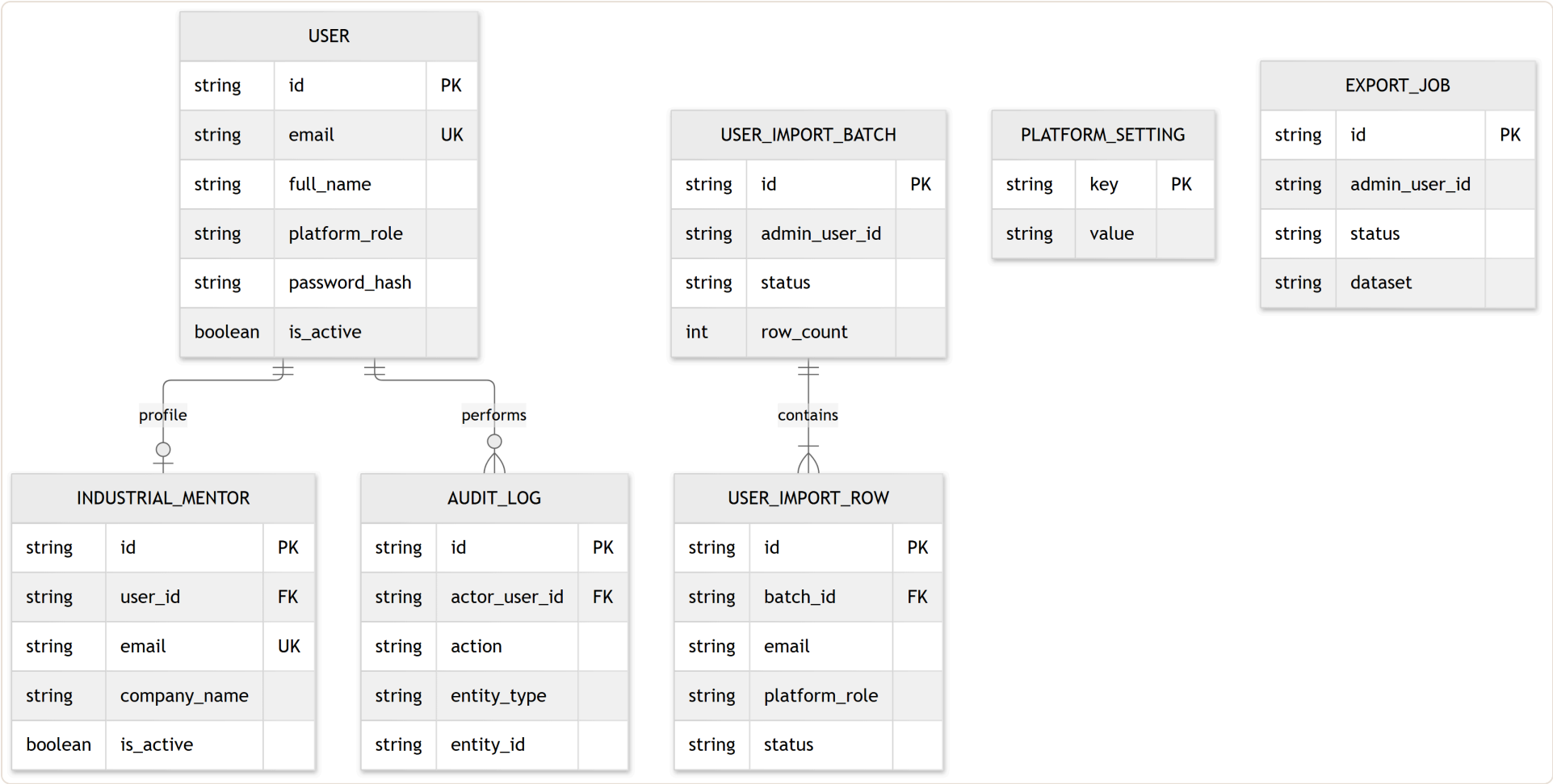
1. High-level domain map

Six bounded contexts: identity, teams, problem statements, mentors, competition stages, and notifications.



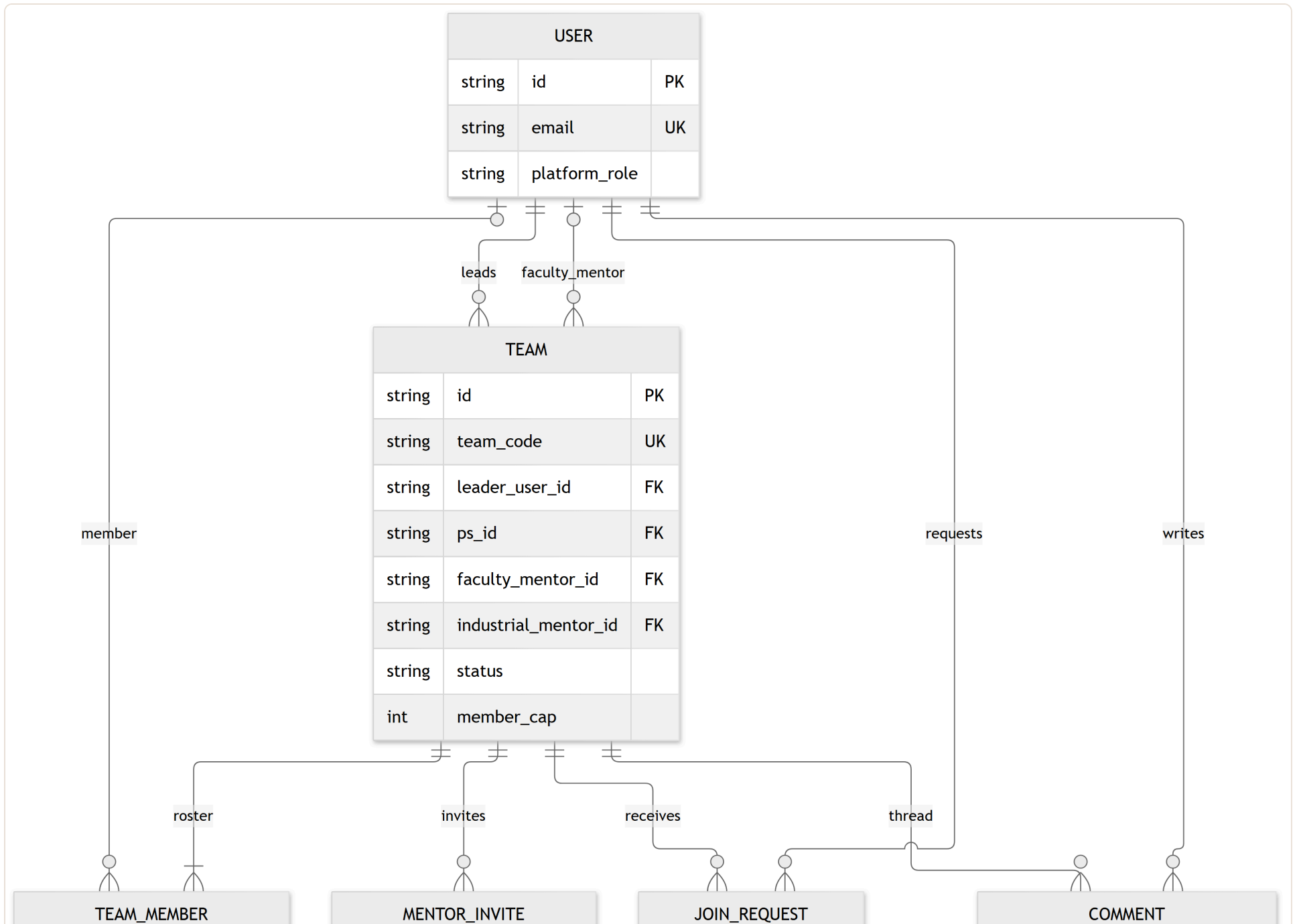
2. Identity and users

Roles: student, institute_mentor, industry_mentor, admin, student_expert. Auth uses password_hash + JWT (no external identity provider).



3. Teams and membership

Teams identified by team_code. Invites tracked in team_members and mentor_invites; email via Resend.



string	id	PK
string	team_id	FK
string	user_id	FK
string	invited_email	
string	invite_status	
datetime	joined_at	

string	id	PK
string	team_id	FK
string	invited_email	
string	mentor_user_id	FK
string	mentor_type	
string	invite_status	

string	id	PK
string	student_id	FK
string	team_id	FK
string	status	



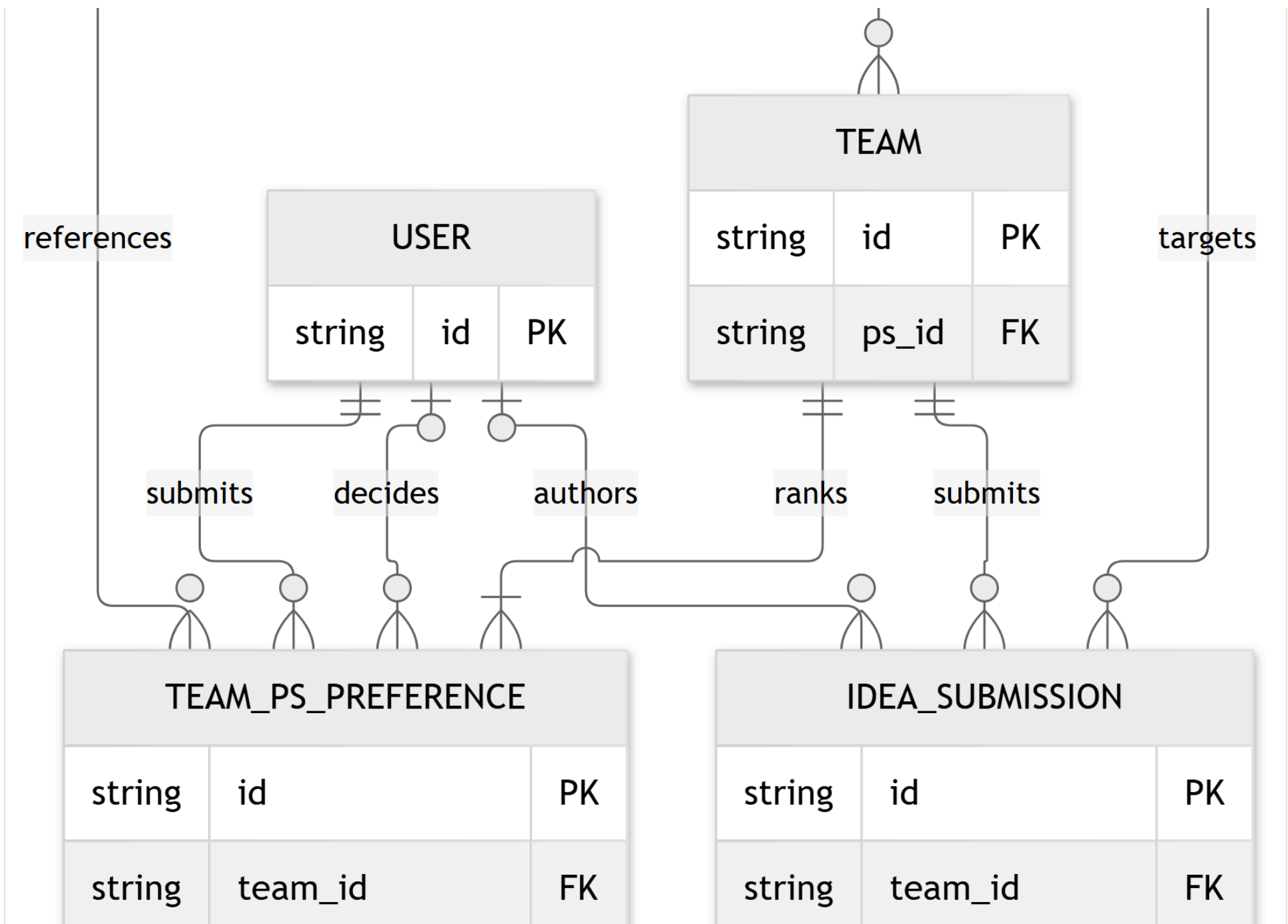
string	id	PK
string	team_id	FK
string	author_user_id	FK
string	parent_comment_id	FK
string	message	

4. Problem statements and ideas

Teams rank PS choices; mentors approve. idea_submissions hold draft/locked ideas per team.

PROBLEM_STATEMENT		
string	id	PK
string	code	UK
string	title	
string	theme	
string	category	
string	organisation	
int	team_cap	



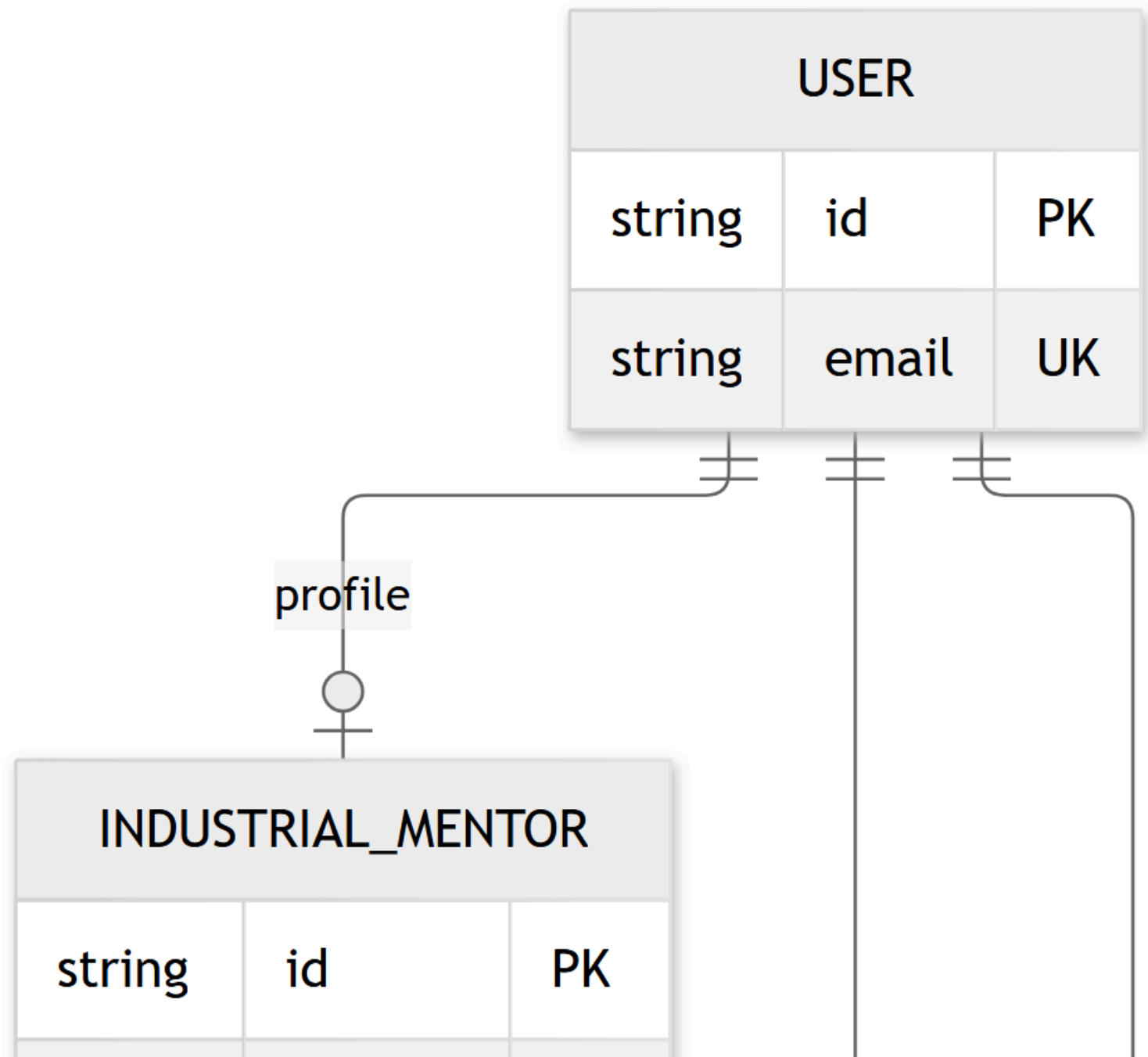


int	rank	
string	ps_id	FK
string	status	
string	submitted_by_id	FK
string	decided_by_id	FK

string	ps_id	FK
string	author_user_id	FK
string	status	
int	version	

5. Mentor allocation

mentor_assignments is the audit trail; faculty and industry mentor slots are denormalized on teams.



string	user_id	FK
string	email	UK

industry_slot



mentor

assigned_by

TEAM

string	id	PK
string	faculty_mentor_id	FK
string	industrial_mentor_id	FK

string	industrial_mentor_id	FK
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history



MENTOR_ASSIGNMENT

string	id	PK
string	team_id	FK
string	mentor_user_id	FK
string	assigned_by_id	FK



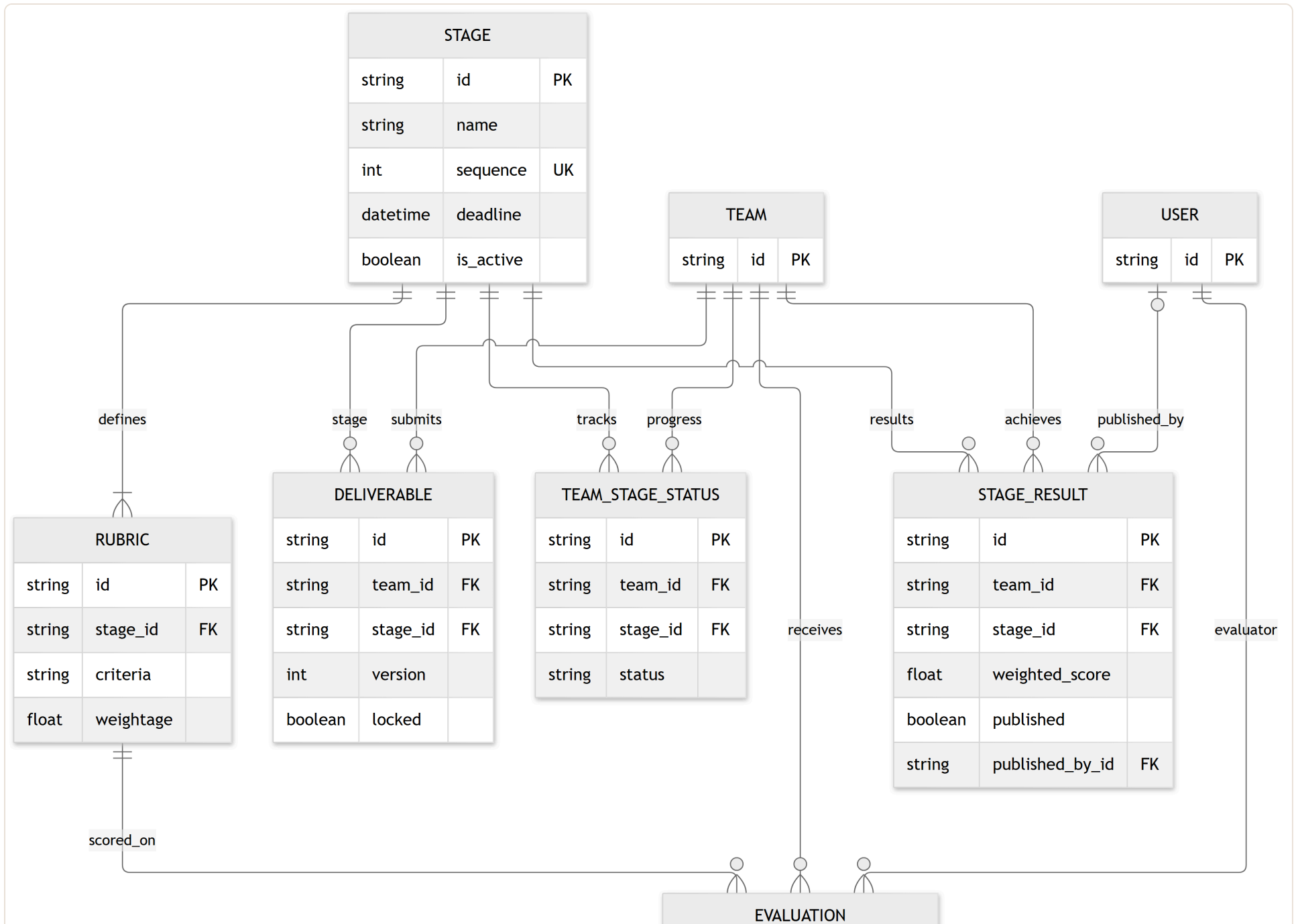
reassign

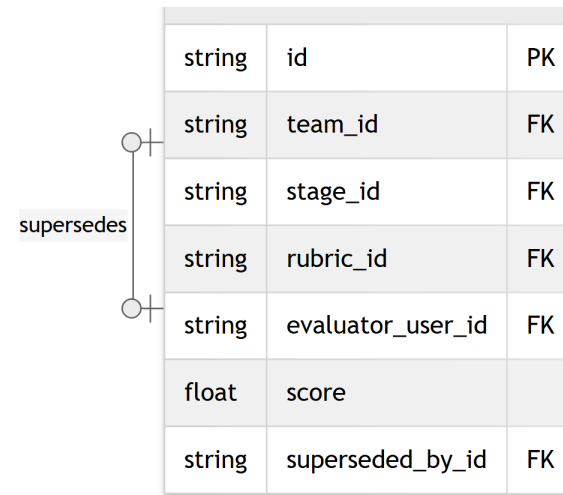


string	industrial_mentor_id	FK
string	mentor_type	
string	assignment_method	
string	reassigned_from_id	FK
boolean	active	

6. Stages, deliverables, and evaluation

Team × stage progress via team_stage_status, deliverables, evaluations, and published stage_results.





7. Notifications and broadcasts

In-app notifications plus notification_log for email delivery tracking.

USER		
string	id	PK
string	email	UK

in_app

email_log

admin_sends

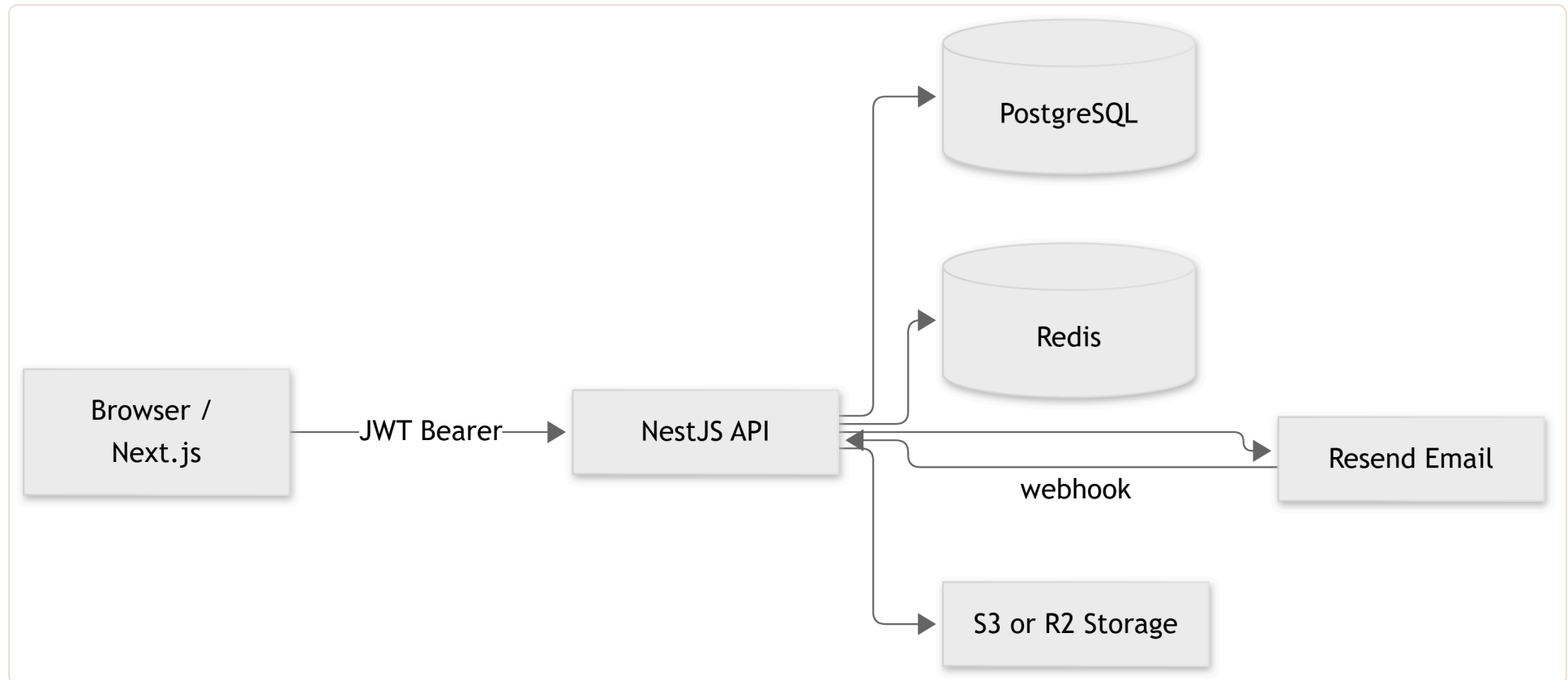
NOTIFICATION		
string	id	PK
string	user_id	FK
string	type	
string	title	
string	body	
datetime	read_at	

NOTIFICATION_LOG		
string	id	PK
string	recipient_user_id	FK
string	recipient_email	
string	template	
string	status	

BROADCAST		
string	id	PK
string	admin_user_id	FK
string	title	
string	body	
int	recipient_count	
datetime	sent_at	

8. External systems

Not stored in PostgreSQL: Redis (OTP, queues), Resend (email), S3/R2 (files).



9. All tables (25)

#	Table	Purpose
1	users	Accounts, roles, credentials
2	teams	SIH squads
3	team_members	Roster and email invites
4	industrial_mentors	Industry mentor profiles
5	problem_statements	Official PS catalog
6	team_ps_preferences	Ranked PS choices
7	idea_submissions	Team idea drafts
8	join_requests	Student join requests
9	mentor_assignments	Mentor allocation history
10	mentor_invites	Team to mentor invites
11	stages	Competition milestones
12	rubrics	Scoring criteria
13	deliverables	Stage uploads
14	team_stage_status	Per-team stage progress
15	evaluations	Mentor scores
16	stage_results	Published results
17	comments	Team threads
18	notifications	In-app alerts
19	notification_log	Email delivery log
20	broadcasts	Admin broadcasts
21	audit_log	Change history
22	export_jobs	Data exports
23	platform_settings	Runtime config

#	Table	Purpose
24	user_import_batches	Bulk import batches
25	user_import_rows	Import batch rows

10. Core relationships

From	To	Card.	FK / notes
User	Team	1:N	leader_user_id
User	Team	0:1	faculty_mentor_id
IndustrialMentor	Team	0:1	industrial_mentor_id
Team	TeamMember	1:N	unique team_id + invited_email
Team	ProblemStatement	N:1	ps_id optional
Team	Stage	N:M	via junction tables
MentorAssignment	MentorAssignment	chain	reassigned_from_id